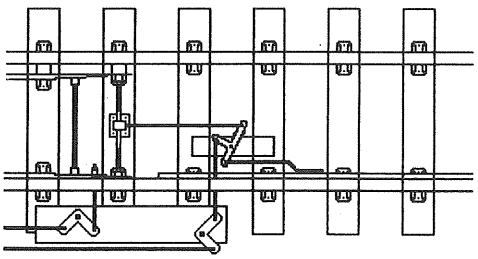
INTRODUCTION TO SIGNALLING

5

POINTS



A.J.R.Rowbotham

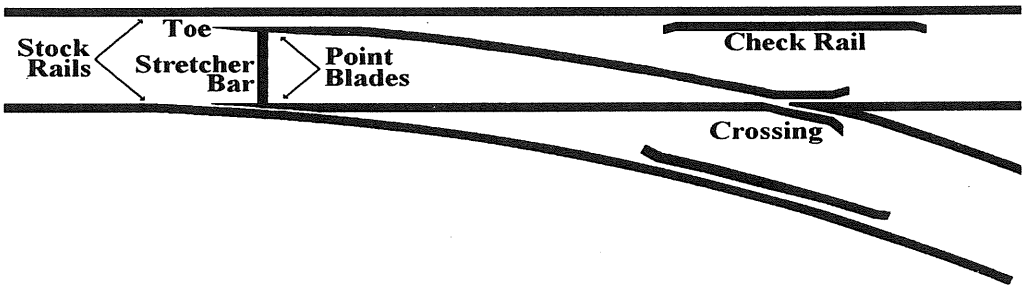
INTRODUCTION

'Points' is the term used to describe the arrangement of rails that allow junctions to be made in the track. The word is usually used in the plural as 'points' or a 'set of points', unless it is being used as an adjective, in which case it is used in the singular.

POINTS AND CROSSINGS

Simple Points

The formation of rails (looking from above) to make a set of points is shown below.



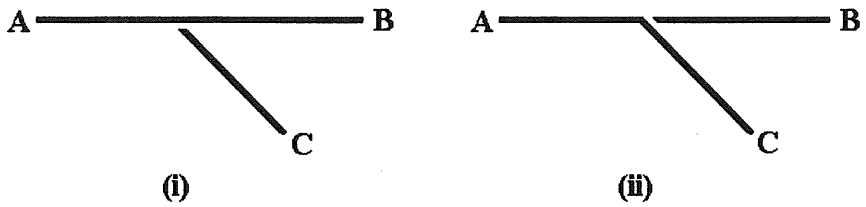
The outside rails which are fixed, are the *stock rails*. The two inside rails, which can move from side to side at one end are the *point blades* (also called *switch blades*). The end that can move is known as the *toe* of the points, and the two blades are held the correct distance apart by a *stretcher bar*. The area where one rail crosses the other is known as *the crossing*. As a wheel rolls over the crossing there is a small area where it is only partly supported. This gap is caused by needing space for the flange of the wheels to pass through the crossing area. As there is a very slight possibility of a derailment at the instant the wheel is unsupported, *check rails* are provided. These are positioned so that the wheel flanges pass between the running rail and the check rail. At the first hint of any problem, the back of the wheel would hit the check rail, which would hold everything in place.

With the point blades positioned as shown above trains may pass along the 'straight' track in either direction. If the two point blades were to be moved so that the upper blade (in the diagram) was touching the straight stock rail, (whilst the lower blade was therefore not touching the curved stock rail), then trains may pass along the 'curved' track in either direction.

Symbols and Nomenclature

The words used to describe the way a pair of points are lying, (or their '*position*'), *Normal* and *Reverse*, were introduced many years ago, and today are rather meaningless. When the decision has been made at the design stage which position is to be termed normal, and which reverse, it is recorded on the track plan. On the track plan, the normal lie of the points is shown by a continuous line of track, whilst the reverse lie is a broken line. It is <u>not</u> possible when standing on the track to determine positively which is normal and which is reverse.

If the points were shown on the track plan as in (i) below, then the points would need to be lying normal for moves between A and B, and would need to be reverse for moves between A and C. However, if the same points happen to have been shown on the track plan as in (ii), then normal setting would be for moves between A and C, and reverse setting for moves between A and B.



A move approaching the point where there is a possibility of a choice of route (such as a move from A to B or C above) is said to be a *facing* move. A move in the opposite direction where there is no choice of route (from B to A, or from C to A, above) is said to be a *trailing* move.

Normal and reverse refer to the direction the points are set , while facing and trailing describe the direction of traffic. These are two independent sets of parameters, there is no correlation between the two. In (i) above the parameters may be summarised:

MOVE	*DESCRIPTION*
A to B	Facing, normal
A to C	Facing, reverse
B to A	Trailing, normal
C to A	Trailing, reverse

Swing Nose Crossing

As mentioned above there are gaps in the rails in the area of the crossing. The narrower the angle of divergence of the points, the longer the gap will be. For high speed lines, junctions are required where the diverging (curved) route may be taken at, say 160 km/h. This means very long point blades and a very narrow angle indeed, a combination that results in an unacceptable gap in the rails at the crossing. To overcome this problem a special crossing, the *swing nose crossing*, has been developed. Here, the V-shaped rail can be pivoted sideways relative to the longitudinal axis of the points, closing up the gap between it and one or other of the point blades.



The moving of the crossing and the point blades must of course be co-ordinated to ensure the position of the blades matches the position of the crossing.

Crossovers

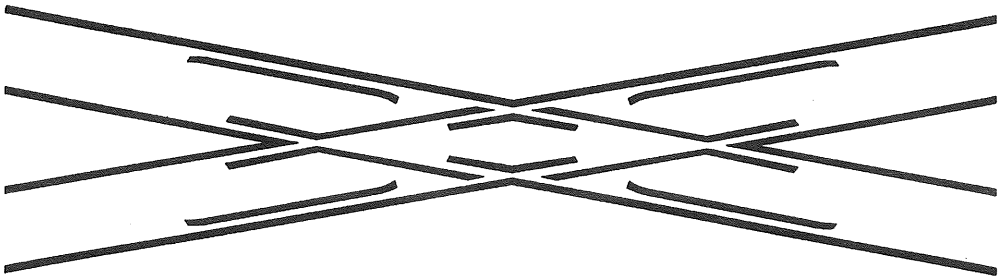
A very common formation of points is a connection between the two parallel lines of a double track. This combination of two points is known as a crossover and it might well be that both points are worked together under one control (*double ended* points). Invariably the normal position of the two points would allow straight through moves on both lines, with both being reverse for a move between the two lines.



A crossover where a train could cross directly from one line to the other is a *facing crossover*. The mirror image, the *trailing crossover* requires the train to reverse to cross over from one line to the other. If both types were to be needed in the same vicinity, but not enough linear space was available, the two types may be superimposed on top of each other, forming a *scissors crossover*. This introduces a new feature of track work - two lines crossing over each other on the level.

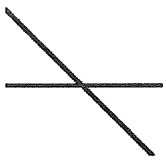
Flat Crossing

The formation of rails to make up a *flat crossing* (or *diamond crossing*, reflecting the shape the rails make in the centre) is shown below. There is no usable connection between the two lines involved.



At each end there is a crossing similar to that found on a set of points, and in the middle are two obtuse crossings. These crossings also have to have gaps to allow the wheel flanges through and have protective check rails, forming a 'K' shape.

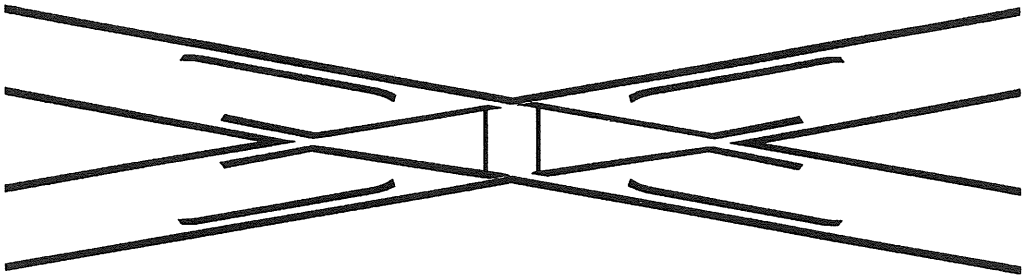
On a track plan the two lines just cross each other at an angle.



As with the points, for high speed junctions the angle of the crossing becomes very narrow and the gaps become unacceptable. The solution is to work the two obtuse crossings as two sets of points - the *switch diamond*.

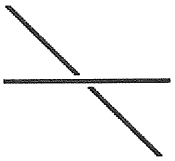
Switch diamond crossings

In the diagram the two sets of point blades are shown with their stretcher bars.



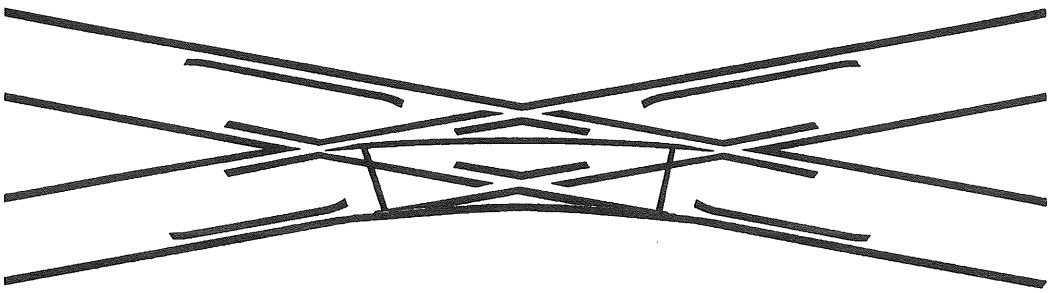
The left hand set of blades are positioned so the lower blade (on the drawing) is touching the lower obtuse angle, whilst the right hand blades are positioned touching the upper obtuse angle. There is thus a path from the upper left track to and from the lower right track, with no gaps at the obtuse angles. For a train to move along the other diagonal path, both sets of blades have to be moved, in opposite directions, to their other position.

On a track plan the usual rules apply, the normal position of the switch diamond is shown as a continuous line, with the reverse position broken. In the diagram the normal position is for 'horizontal' moves, and the reverse is for 'diagonal' moves.



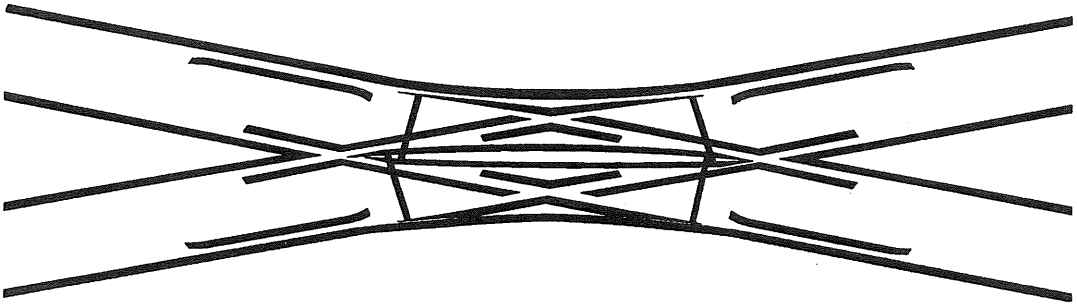
Slips

It is possible to add rails between the outer crossings to allow movement around one of the corners. Such an arrangement is known as a *single slip* (you can 'slip' round one of the corners).



In the diagram it is possible to make a move between the lower two ends. It would be equally possible to allow movements between the upper two ends instead with a mirror image of this arrangement.

If both moves are required the track work becomes complex and the combination is known as a *double slip*.



It is possible to have a double slip with switch diamonds but this combination would be very expensive and very rare.

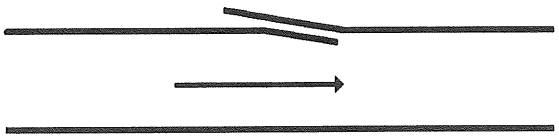
CATCH AND TRAP POINTS

It is sometimes necessary to deliberately derail a train to protect a second train from a possible collision. Two different methods, *catch points* and *trap points* are used in different circumstances.

Catch Points

Since the 1880s it has been a requirement for passenger carrying vehicles to have continuous brakes, that is, every carriage had brakes operated through a brake pipe from the engine. If a coupling failed so the train divided into two, the brake pipe in both halves would be open to the atmosphere and the brakes automatically applied. This was not the case for goods wagons, however, and until comparatively recently individual wagons did not have brakes. Braking the train relied solely on the brakes on the locomotive. If a coupling on a goods train failed, brakes would not be automatically applied. and indeed the driver might not even be aware of a problem. If the train was ascending a hill at the time, there would be a distinct danger of the rear part of the train accelerating backwards down the hill and colliding with a following train.

Catch points used to be installed on long hills to derail anything running backwards against the normal direction of traffic, and are not controlled by the signaller. The spring catch point consists only of one point blade with no crossing. The blade is normally open, in the derailing position.



In the diagram the normal direction of travel is from left to right on an ascending gradient. Each wheel of a train travelling in the correct direction pushes the point blade closed against a spring. Any vehicle running back down the hill from right to left will be derailed at the catch point.

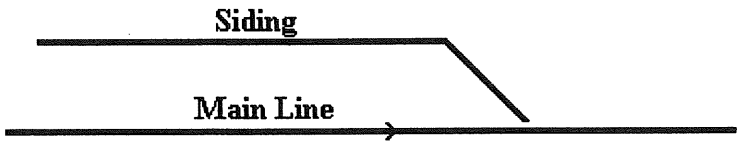
The symbol used on track plans incorporates a letter 'S' to indicate the spring mechanism. The normal (in the signalling sense) position of the points is the derailing position, and trains travelling in the normal direction (in the traffic sense) are making trailing moves through them.



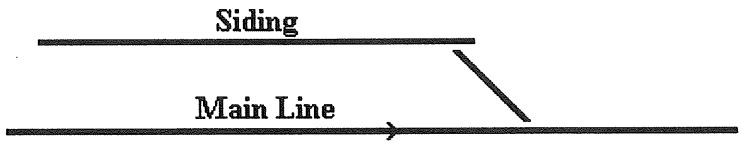
As all vehicles are now fitted with continuous brakes, catch points are not normally replaced when track is relaid. The numbers found are consequently on a sharp decline.

Trap Points

Trap points are installed in particular circumstances to provide protection from trains that might have inadvertently passed a signal. They are controlled by the signaller like any other point. Consider a siding connected to the main line by a trailing connection:

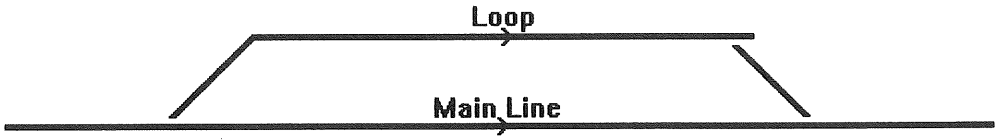


There would be a signal at the exit of the siding to control trains wanting to regain the main line. However, in the siding, wagons might well be shunted about and it might well happen that a wagon ran out of the siding out of control. If this were to happen then a train moving along the main line could be hit. To prevent this happening a second point end, making up a crossover, is installed at the exit, leading to nowhere.



If a train is moving along the main line both point ends will be normal, so if anything runs out of the siding it will just fall off the end of the line and be derailed. For a train to legitimately enter or leave the siding both ends of the double-ender would be reverse.

Similar arguments apply at the exit of a loop line, to perhaps let fast trains pass slower trains.

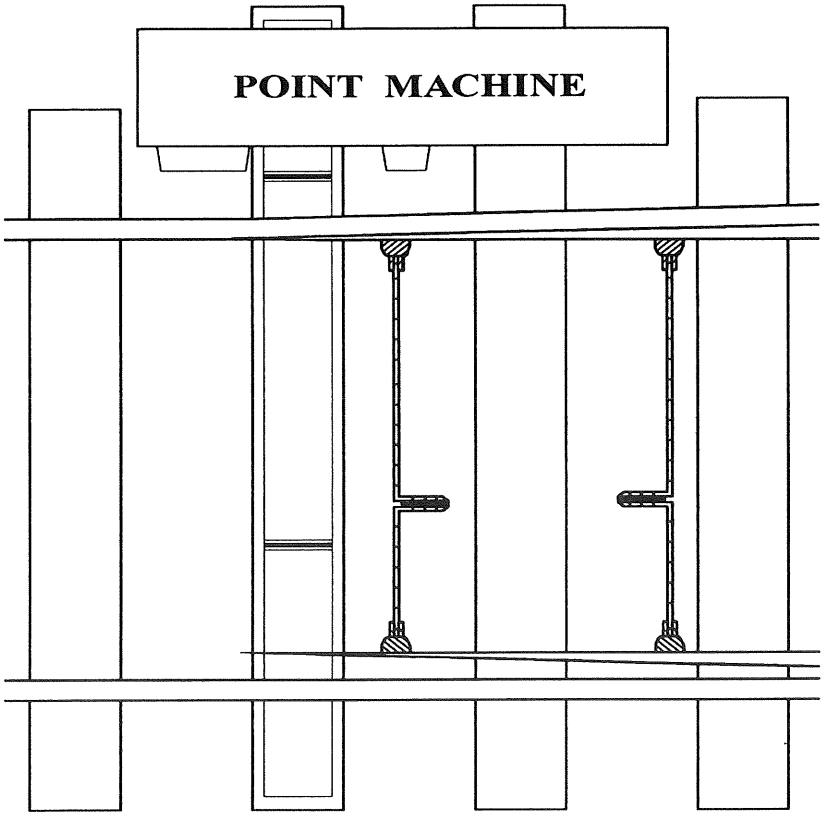


Any train moving along the loop that failed to stop at the exit signal would be prevented from colliding with a train on the main line by trap points installed at the exit of the loop.

POINT MACHINES

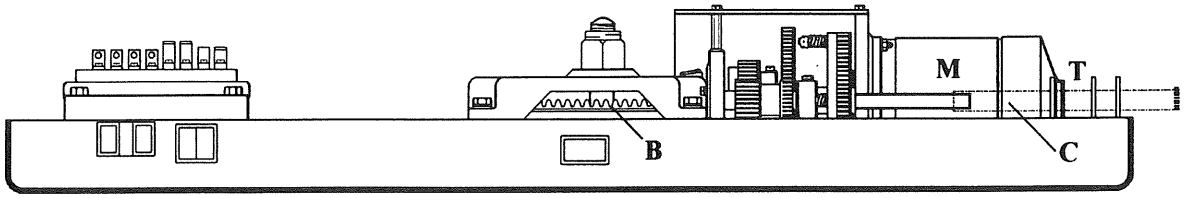
The position of the point blades is controlled by a *point machine*, which drives the blades to the correct alignment. The term conventionally refers to a machine incorporating an electric motor. A newer development, the *clamp lock*, incorporates hydraulic power and is described later.

The point machine is fixed to extra long bearers ('*extended timbers*') adjacent to the toes of the points.



It is held in place relative to the stock rails by a *sole plate*, a metal plate bolted on the top of the sleeper under the toe of the point blades. The points are supplied by the Civil Engineer fitted with at least two stretchers. The stretchers and the sole plate normally need insulation to electrically separate the two stock rails and the machine (shown solid black above).

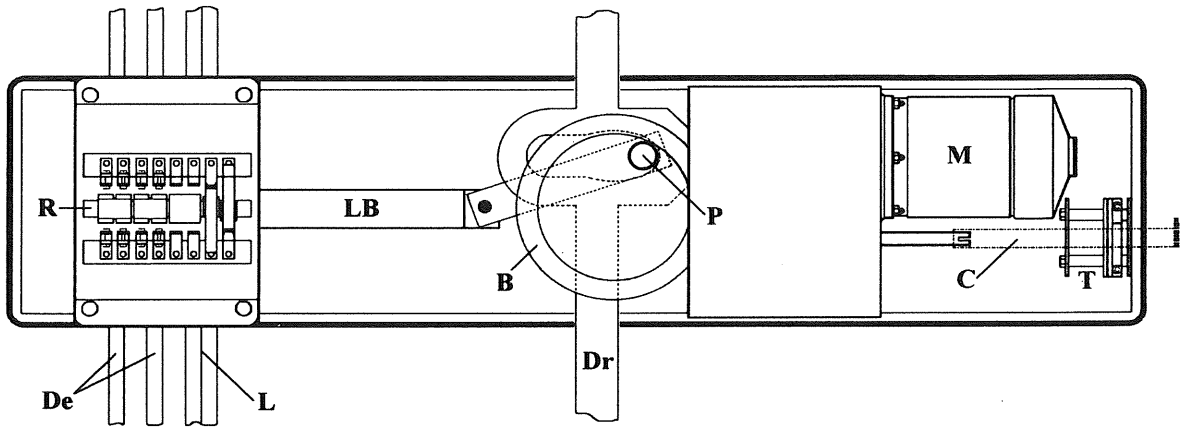
A point machine with its cover removed is shown below.



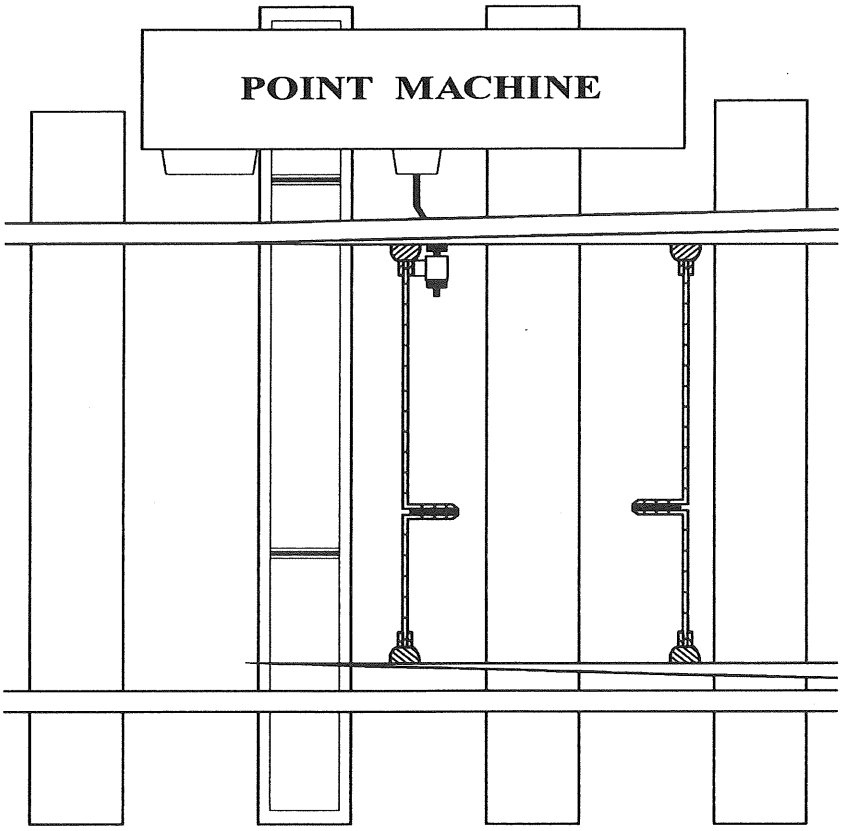
An electric motor (M) drives, through a gear train and clutch, a large horizontal bevel gear (B). The clutch is incorporated to protect the motor if the blades are obstructed. (Coal from steam engines jamming the blades used to be a very common occurrence). Further protection

is provided by the interlocking with a timer that only allows the motor to be powered for up to eight seconds.

The bevel gear has a pin (P) fitted which engages in the 'foot print' shape machined out of the *drive rod* (Dr). (In the plan view below the cover over the bevel gear has been removed.)



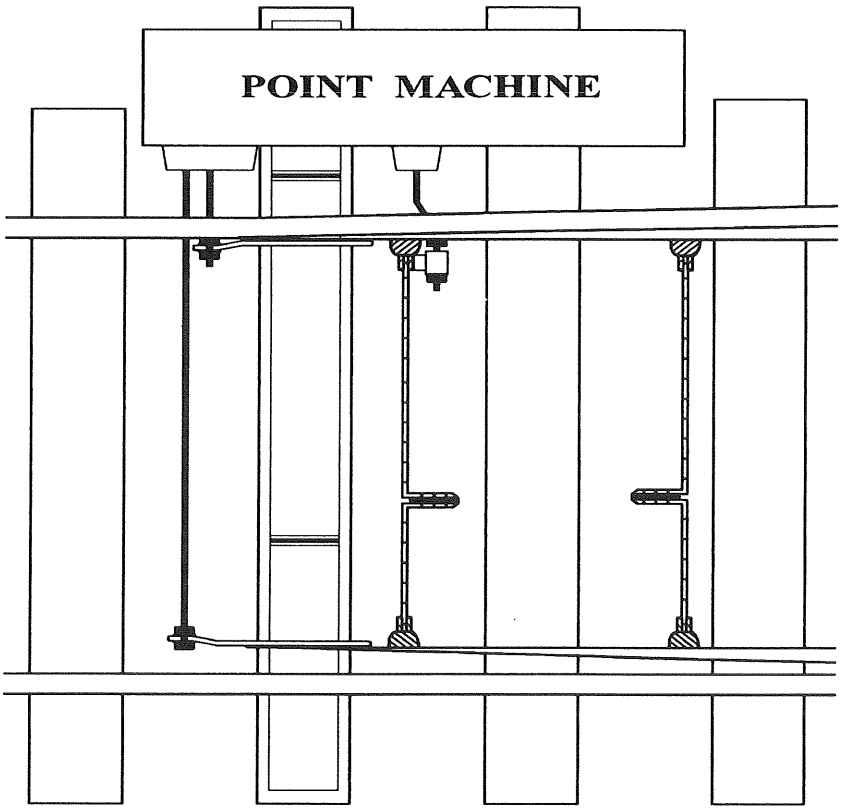
As the bevel gear rotates about three-quarters of one revolution (anticlockwise from the position shown), the rotary motion is converted to linear motion of the drive rod. This drive rod is connected to one of the stretchers to position the blades.



If the motor fails, the machine may be operated manually by inserting a crank handle (C), (partly shown, chain dotted) in the trunnions (T) at the end of the machine, adjacent to the motor. The action of inserting the crank handle electrically isolates the motor so no injury can

occur to the person using the handle if the motor was to suddenly rotate. When the handle is removed, a plunger has to be operated to reinstate the motor connection.

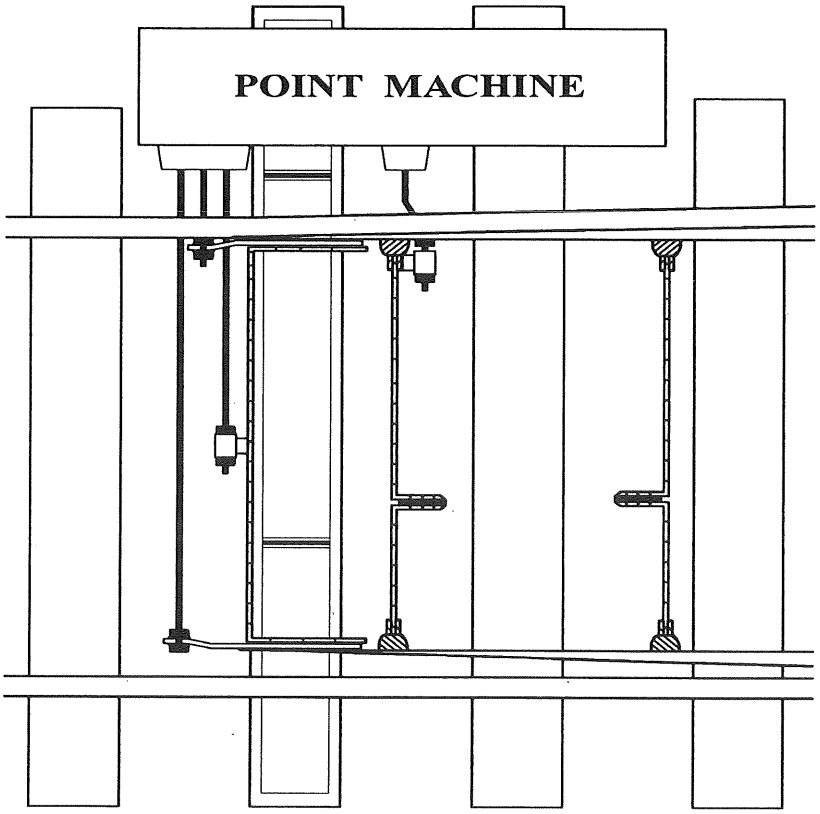
It is essential for the interlocking to know the position of the blades as signals must not show a proceed aspect unless the points are proved to be set the correct way. An individual mechanical connection, the *detector rod* (De), is taken from an extension piece on each point blade into the machine to work accurately set *detection* contacts. Two parallel banks of contacts may be seen at the left hand end of the diagrams above. A rocker arm (R) between the two banks (shown in the mid-position) opens and closes the contacts. The four contacts in each bank nearest the end of the machine are used to control the detection circuits back to the interlocking. The other four in each bank are heavy duty and are used for internal switching within the machine. The innermost contacts remain closed whilst the point blades are moving, and are broken when the correct position is reached. The rocker arm is operated by the detector rods so that the normal or reverse contacts are only operated when the appropriate blade is physically in the closed position and the other is open. Misplacement of either blade by a few millimetres is enough to prevent the contacts operating (and thus preventing the signals from showing a proceed aspect).



Since the early days of railways there has been a fear of the blades actually moving while a train is passing over them in the facing direction. This would result in a derailment and loss of life if it occurred at high speed. Trains trailing through points set in the wrong direction would not normally be derailed, although the points would be damaged. Points operated mechanically from a lever frame by rodding had one lever to move them and a second lever to work a mechanical lock mounted in the four foot if facing moves took place. A mechanical

lock is built into the machine and is connected physically to the blades by a *lock rod* (L), which is connected to a lock stretcher bar. This extra stretcher bar is provided by the Signal Engineer and is attached to (and insulated from) the extension pieces. The locking is achieved within the machine by a bar (LB) parallel with the axis of the machine. This is connected by a short pivoted bar to the pin fitted to the bevel gear. At the other end, under the detector contacts, a part of this bar passes as a close fit through one of two slots cut in the lock rod (one slot when locking the points normal, the other slot when locking the points reverse). The shape of the 'foot-print' cut out is such that when the motor starts turning, firstly the internal lock is withdrawn from one slot, then the blades are moved across, and finally the internal lock is remade in the other slot. The detection contacts only make up when the detection is made up and the locking is effective.

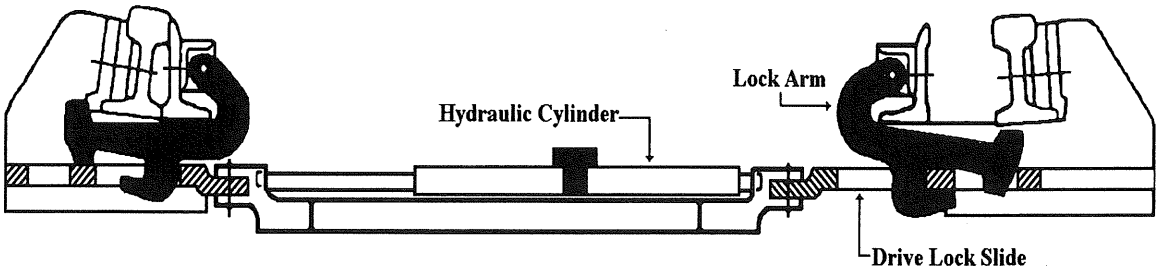
In total there are four mechanical connections between the machine and the blades - the drive, the lock and the detection for each blade.



It must be noted that the detection contacts are in fact monitoring the position of the blades relative to the point machine rather than the stock rails that they are supposed to be touching. The clamp lock mechanism goes some way to avoid this problem.

CLAMP LOCKS

The clamp lock mechanism is designed to hold the closed point blade positively locked against its stock rail. In the diagram below, looking at a cross-section across the track, a Lock Arm is pivoted



from each point blade (shown black in the diagram). These are operated by a Drive Lock Slide under each rail. The Lock Slides (shown crosshatched in the diagram) have slots cut in them, into which the Lock Arm can fall. The two Lock Slides are connected together, and are driven to the normal or reverse positions by a hydraulic cylinder mounted in the four foot. The power is provided by an hydraulic pump by the side of the track, which together with the cylinder, is controlled from the interlocking.

In the diagram the left hand blade is shown closed, and the Lock Arm is held in such a position by the Lock Slide that the blade is clamped against the stock rail (by the hammer-head shape at the left hand end of the Arm). When movement is required, the cylinder starts driving both Lock Slides to the right. The first thing that happens is the left hand Lock Arm drops into the left hand slot of the left hand Lock Slide, unlocking the blade. Further movement of the Lock Slides starts moving both blades to the right, and the last action is the right hand Lock Slide forcing the right hand Lock Arm upwards to lock the right hand blade against its stock rail.

The detection is carried out by commercial heavy-duty micro-switches mounted in the clamp lock mechanism bolted to each stock rail. These not only detect that the closed blade is clamped in the correct position, but also that the open blade is positioned correctly, that is at least at the minimum opening.

When clamp locks were introduced it was thought that they would become the standard for operating points. The advantages stated included that

- extended timbers were not needed.
- there were no extremely heavy weights to contend with.
- long drive, lock and detector rods, which are vulnerable to tamping machine damage, are not required.

However, after a number of years experience they seem now to be out of fashion. They have not proved to be very reliable, one reason being the detectors bolted directly to the rails, with every train passing giving them a good shake. It has also been found that the drilling of the rails for the attachment of the equipment has to be very accurate, and that the permanent way in the area has to be maintained to a high standard.